Worksheet #3

MAA 4402 — July 5, 2023

1. Use the definitions $\cos(z) = \frac{e^{iz}+e^{-iz}}{2}$ and $\sin(z) = \frac{e^{iz}-e^{-iz}}{2i}$ to prove that for each $z \in \mathbb{C}$
$$\cos(z)^2 + \sin(z)^2 = 1$$

2. Use the definition of $\cos(z)$ to show that $(\cos(z))' = -\sin(z)$

3. Determine all points where the following functions are differentiable and compute the derivative at these points.

   a. $f(z) = iz + 2$

   b. $f(z) = z - \overline{z}$

   c. $f(z) = z\mathsf{Im}(z)$

   d. $f(z) = 1/z$.

   e. $f(z) = |z|^2$

4. Let $f(x+iy) = u(x,y) + iv(x,y)$ be analytic. Use the Cauchy Riemann equations to show that $u_{xx} + u_{yy} = 0$ and $v_{xx} + v_{yy} = 0$.